

The diagram illustrates the circuit for the MB506 divider, which is used to convert a 3V signal to a 5V signal for the Arduino Pro Mini. The circuit is divided into two main sections: the MB506 divider and the Arduino Pro Mini connection.

PREDIVISEUR MB506 (division fixe 1/64)

The MB506 divider (U3) is a fixed 1/64 divider. It is powered by a +5V supply (pin 3 and 6) and has a GND connection (pin 5). The input signal (RF_IN) is connected to pin 1, and the output signal (D5_SIG) is connected to pin 2. The output signal is also connected to the D5_SIG pin of the Arduino Pro Mini (A1).

MISE EN FORME 3V -> D5

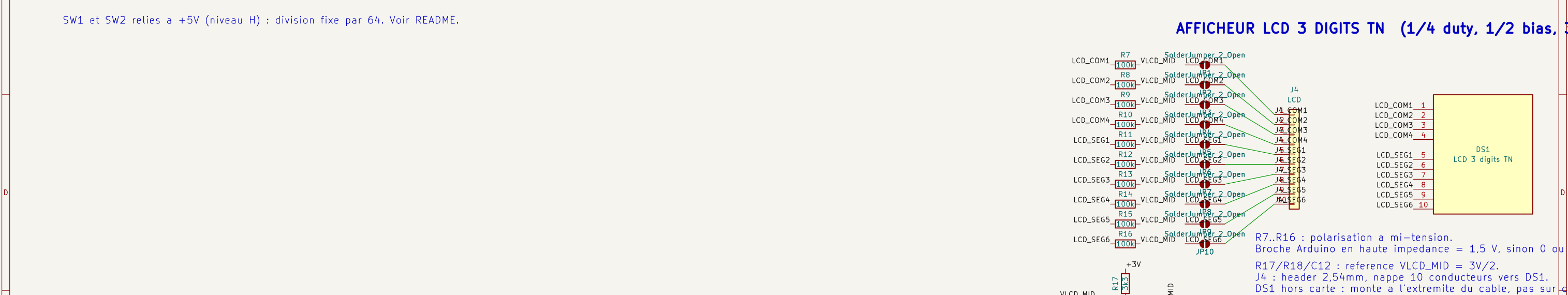
The output signal (D5_SIG) is connected to the D5_SIG pin of the Arduino Pro Mini (A1). The output signal is also connected to the D5_SIG pin of the Arduino Pro Mini (A1).

ARDUINO PRO MINI (alimente en 3V)

The Arduino Pro Mini (A1) is powered by a 3V supply (pin 21) and has a GND connection (pin 20). The output signal (D5_SIG) is connected to the D5_SIG pin (pin 8) and the D5_SIG pin (pin 12). The output signal is also connected to the D5_SIG pin (pin 17) and the D5_SIG pin (pin 18).

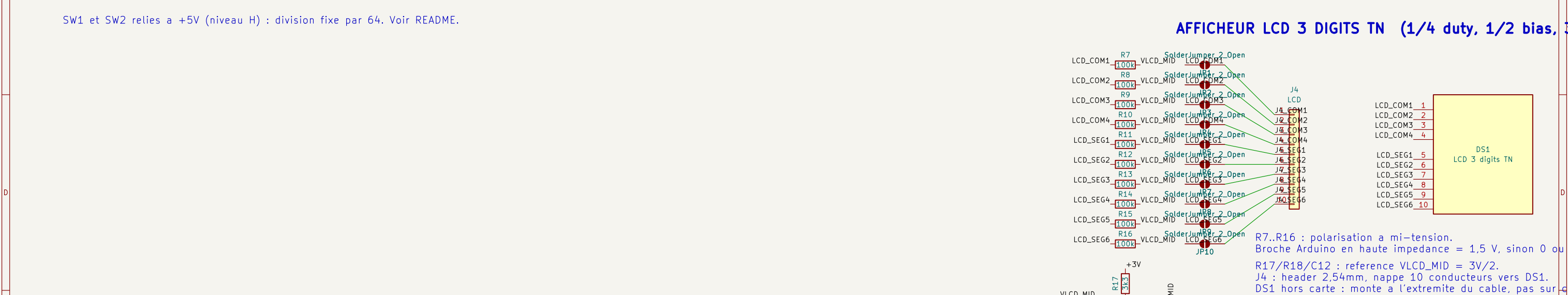
The diagram illustrates the circuit for the MB506 pre-amplifier and its connection to an Arduino Pro Mini. The circuit is divided into three main sections:

- PREDIVISEUR MB506 (division fixe 1/64):** This section shows the MB506 chip (U3) configured as a pre-amplifier. It has two input ports: RF_IN and RF_J2_IN. The output is connected to a 2SC5551AE-TD-E transistor (Q1). The circuit includes several passive components: resistors R3 (151R), R4 (3k3), R6 (470R), R19 (2k2), and R20 (470R); capacitors C7 (1n), C8 (100n), C9 (100n), C10 (100n), and C11 (100n). The MB506 chip is powered by +5V and GND.
- MISE EN FORME 3V -> D5:** This section shows the transistor Q1 (2SC5551AE-TD-E) configured as an emitter follower. The output of the transistor is connected to the D5_SIG pin of the Arduino Pro Mini. The transistor is powered by +3V and GND.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini board (A1) connected to the output of the transistor. The board is powered by +3V and GND. The output of the transistor is connected to the D5_SIG pin of the Arduino Pro Mini.



The diagram illustrates the circuit for the MB506 pre-amplifier and its connection to an Arduino Pro Mini. The circuit is divided into three main sections:

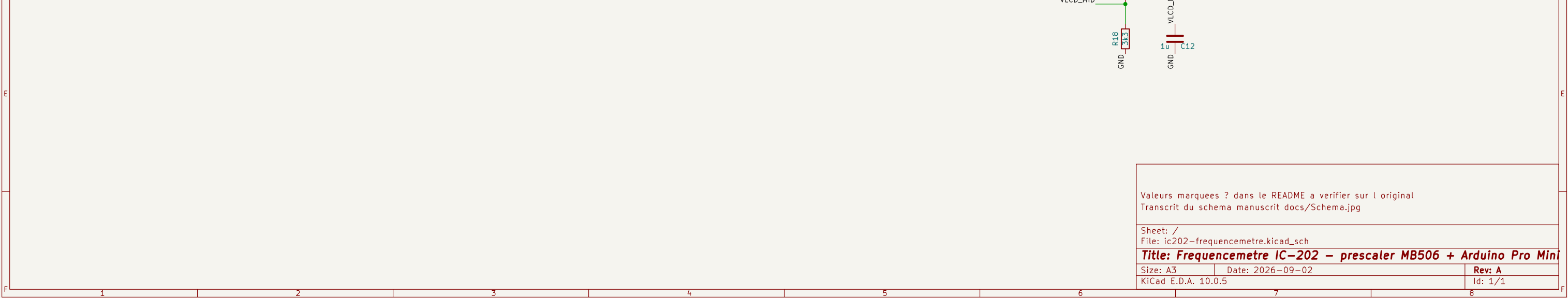
- PREDIVISEUR MB506 (division fixe 1/64):** This section shows the MB506 chip (U3) configured as a pre-amplifier. It has two input ports: RF_IN (pin 1) and MB506_IN (pin 2). The output is MB506_OUT (pin 7). The chip is powered by +5V (pins 3 and 6) and grounded (pins 4, 5, and 8). Biasing is provided by a 15k resistor (R3) and a 1nF capacitor (C7) to the RF_IN, and a 100nF capacitor (C8) and a 470k resistor (R20) to the MB506_IN. A 100nF capacitor (C9) is connected to the +5V supply.
- MISE EN FORME 3V -> D5:** This section shows the signal path from the MB506_OUT to the D5 pin of the Arduino Pro Mini. The signal passes through a 100nF capacitor (C10) and a 3k resistor (R4) to the D5_SIG pin (pin 8) of the Arduino. A 2SC5551AE-TD-E transistor (Q1) is used as a buffer, with its base connected to the signal line and its emitter to ground.
- ARDUINO PRO MINI (alimente en 3V):** This section shows the Arduino Pro Mini (A1) powered by a 3V battery (C11). The power pins are 24 (GND), 23 (GND), 22 (GND), 21 (+3V), 20 (+3V), 19 (+3V), 18 (+3V), 17 (+3V), 16 (+3V), 15 (LCD_SEG6), 14 (LCD_SEG5), and 13 (LCD_SEG4). The signal pins are 1 (GND), 2 (GND), 3 (GND), 4 (GND), 5 (LCD_COM1), 6 (LCD_COM2), 7 (LCD_COM3), 8 (D5_SIG), 9 (LCD_COM4), 10 (LCD_SEG1), 11 (LCD_SEG2), and 12 (LCD_SEG3).



SW1 et SW2 relies a +5V (niveau H) : division fixe par 64. Voir README.

AFFICHEUR LCD 3 DIGITS TN (1/4 duty, 1/2 bias, 1/20 V)

R7 VLCD_MID
 R8 VLCD_MID
 R9 VLCD_MID
 R10 VLCD_MID
 R11 VLCD_MID
 R12 VLCD_MID
 R13 VLCD_MID
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 R19 VLCD_MID
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 R21 VLCD_MID
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